

$$y[n]=\frac{1}{a_0}\left(\sum_{i=0}^N b_i x[n-i]-\sum_{j=1}^M a_j y[n-j]\right)$$

$$H(z)=\frac{b_0+b_1z^{-1}+b_2z^{-2}}{a_0+a_1z^{-1}+a_2z^{-2}}$$